Issue 128 Exact-D8 HPC Sprint

Interim Technical Supplement

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Abstract

This supplement records a six-hour, proof-directed HPC sprint following the machine-certified Issue 128 result. The accepted claim remains frozen at $11791/2911 = 4.050498110614909\ldots$; no fivefold result is asserted here. The sprint computes the order-eight coefficient of the exact right generator as 31 independently checkable stage shards, runs the accepted deep verifier independently, and ranks 24 matching-color orders in an explicitly untrusted discovery path. Every production artifact is atomic, hash-addressed, and bound to a source commit. This is an interim execution report: the final claim gate remains closed until all shards are reduced and a complete global error ledger passes both fast and deep verification.

1 Frozen result and research gate

For the periodic 12×12 isotropic Heisenberg benchmark at $T = 1$ and global operator-norm tolerance 10^{-6}, the certified fourth-order Suzuki result accepts $r = 97$ and rejects $r = 96$. With $G(r) = 30r + 1$ merged group exponentials, the exact resource ratio is

$$\frac{G(393)}{G(97)} = \frac{11791}{2911} = 4.050498110614909\ldots.$$

Fivefold arithmetic would require $r \leq 78$, but this arithmetic is not an error proof. The present experiment targets one missing ingredient: a concrete exact order-eight right-generator coefficient map. A new multiplier may be reported only if a complete D4–D8-plus global ledger at the proposed step count is at most 10^{-6} and independent verification succeeds.

2 Exact sharding algorithm

For stages $V_j(t) = \exp(-ia_j t H_{c_j})$ in the fixed 31-stage formula, linearity gives a decomposition of the right generator into 31 contributions. Each array task starts from $a_j H_{c_j}$ and conjugates it through all later stages, truncating only the formal series degree at eight. Coefficients live exactly in $\mathbb{Q}[\alpha]/(\alpha^3 - 4)$; Pauli operators are canonicalized under the 2×2 translation cell before serialization.

The shard format does not expose process-local registry bit indices. Every Pauli string is encoded by physical integer coordinates and labels. Canonical JSON and gzip with fixed metadata make repeated output byte-deterministic. The manifest records stage, order, timestamps, term counts, peak RSS, source commit, dirty flag, and SHA-256. The reducer rejects a missing stage, wrong formula/order, mixed source state, corrupt hash, or duplicate stage. It also requires forward and reverse exact reduction to produce identical maps.

For a cubic coefficient $c = a_0 + a_1\alpha + a_2\alpha^2$, an outward rational interval for $\alpha = \sqrt[3]{4}$ encloses c. Summing the larger endpoint magnitudes yields a rigorous Pauli-ℓ_1 upper bound. This bound is a screening ingredient; it is not by itself the global $r = 78$ certificate.

3 Execution contract

The production source is commit `e65c95f4ae76dc1a78e81715cac21cfd7a81dbf8`. Jobs run on the `xhacnormalb` CPU partition under account `giggleliu` and QOS `user_acamtw70yu`. GPU partitions are intentionally excluded because the workload is sparse Python integer/rational algebra.

Role	Slurm job	Per-task allocation	Limit
Exact D8 array (31 stages)	23044178	17 CPU, 64 GiB	4:30
Independent verification	23044196	9 CPU, 32 GiB	4:30
Exact reducer	23044212	26 CPU, 96 GiB	1:15
Order screen (24 permutations)	23044214	5 CPU, 16 GiB	1:00

Table 1: Submitted Slurm work. The reducer has an `afterok` dependency on the D8 array.

The array first passed `sbatch -test-only`. A four-cell remote smoke then completed the lightest stages and produced matching deterministic artifacts. The partition's 3931 MiB-per-CPU memory rule was discovered by the test-only gate and reflected in all allocations. Once production started, the voluntary array throttle was raised from 16 to 31; Slurm and QOS retained authority over placement and admission.

4 Verification evidence at launch

The local proof-preservation gate reported 93 passing tests and 11 deselected slow tests in 41.12 seconds. A separate deep verifier run returned `valid=true` and `deep_proof_regenerated=true`, reproduced 393 $\to$ 97, and returned the exact ratio `11791/2911`. At the first full-concurrency snapshot, the D8 run contained 28 running manifests, three complete manifests, and zero failures. The light-stage completion times were approximately 5.61 seconds (stage 28), 0.171 seconds (stage 29), and 0.024 seconds (stage 30). These timings validate the transport and serialization path; they do not predict the much heavier early stages.

The matching-order array is a separate discovery lane. Its JSON records `trusted=false` and states that its purpose is discovery ranking. A favorable ranking can nominate a formula for exact reconstruction but cannot enter the certificate directly.

5 Final adjudication protocol

The outcome will be classified as follows.

1. If all 31 shards and exact reduction succeed, insert the resulting D8 upper bound into a complete $r = 78$ ledger and rerun fast/deep checks.
2. If any task reaches OOM or walltime, retain successful shards and scheduler evidence, retry only missing cells if time remains, and make no inference from a partial sum.
3. If the complete ledger is at most 10^{-6}, prepare a new independently bound certificate before describing a fivefold result.
4. Otherwise publish the strict negative result or remaining numerical gap, while keeping the certified 4.050498-fold result unchanged.

Interim status. This document records a running experiment, not a new performance claim. The accepted result remains 4.050498-fold until the final gate above is closed.